

Лабораторная работа № 9

Настройка IMAP-сервера

Баранов Никита Дмитриевич

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Содержание I

- 1 Информация
- 2 Цель и задачи
- 3 Ход работы
- 4 Результаты

Раздел 1

Информация

- Баранов Никита Дмитриевич



- Баранов Никита Дмитриевич
- Студент РУДН



- Баранов Никита Дмитриевич
- Студент РУДН
- 1132242977@rudn.ru



Раздел 2

Цель и задачи

- Получить навыки установки и настройки Dovecot для доступа к почте по IMAP.

- Установлены Dovecot и telnet; заданы механизмы аутентификации и `mail_location=maildir:~/Maildir`.

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- После создания `Maildir` письма показаны утилитой `s-nail`.
- Сеанс `telnet` на порту 143 подтвердил аутентификацию, `LIST`, `SELECT`, `FETCH`, `STORE` и `LOGOUT`.

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- После создания Maildir письма показаны утилитой `s-nail`.
- Сеанс telnet на порту 143 подтвердил аутентификацию, LIST, SELECT, FETCH, STORE и LOGOUT.
- Письма успешно удаляются через IMAP-флаг Deleted и EXPUNGE.

Раздел 3

Ход работы

Установка Dovecot

- На сервер установлен Dovecot, включены IMAP/POP3-службы и проверен статус процесса.

```
root@server:~  
root@server ~]# dnf -y install dovecot telnet  
Last metadata expiration check: 0:22:32 ago on Sun 06 Sep 2026 08:51:38 AM UTC.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
dovecot	x86_64	1:2.3.21-19.el10_2.1	appstream	4.8 M
telnet	x86_64	1:0.17-94.el10	appstream	62 k

```
=====
```

Transaction Summary

Install 2 Packages

Total download size: 4.8 M
Installed size: 18 M
Downloading Packages:

(1/2): telnet-0.17-94.el10.x86_64.rpm	306 kB/s	62 kB	00:00
(2/2): dovecot-2.3.21-19.el10_2.1.x86_64.rpm	7.2 MB/s	4.8 MB	00:00

Total	4.7 MB/s	4.8 MB	00:01

```
-----  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      :  
  Installing     : telnet-1:0.17-94.el10.x86_64 1/1  
  Running scriptlet: dovecot-1:2.3.21-19.el10_2.1.x86_64 2/2  
  Installing     : dovecot-1:2.3.21-19.el10_2.1.x86_64 2/2  
  Running scriptlet: dovecot-1:2.3.21-19.el10_2.1.x86_64 2/2  
Installed:  
  dovecot-1:2.3.21-19.el10_2.1.x86_64 telnet-1:0.17-94.el10.x86_64  
complete!
```

Параметры протоколов

- В конфигурации включены imap и pop3, настроены механизмы обычной системной аутентификации.

```
root@server:~  
[root@server ~]# postconf -e 'home_mailbox = Maildir/'  
[root@server ~]# firewall-cmd --get-services  
firewall-cmd --add-service=pop3 --permanent  
firewall-cmd --add-service=pop3s --permanent  
firewall-cmd --add-service=imap --permanent  
firewall-cmd --add-service=imaps --permanent  
firewall-cmd --reload  
firewall-cmd --list-services  
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602  
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo  
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export  
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat  
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls d  
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forewa  
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli  
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i  
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib  
ana klogind kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub  
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler  
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net  
work llmn llmn-client llmn-tcp llmn-udp managesieve matrix mdns memcache minecraft minidlna mnpd mon  
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi  
os-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag  
econsole ovirt-vmconsole plex pmcd pmpoxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p  
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netdrv ptp pulseaudio puppetmaster quassel radi  
us radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client  
samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls  
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-t  
ransfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn synct  
hing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terrari  
a tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsu vnc-server vrpp warpinator  
wbem-http wbm-http wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-di  
scovery-udp wsd wsd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent  
zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier  
success  
success  
success  
success  
success  
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom  
[root@server ~]# restorecon -vR /etc
```

- Параметр `mail_location` указывает на `~/Maildir`.

The screenshot shows a web browser window on the left and a terminal window on the right. The browser window displays a document titled "009-imap.pdf" with a search bar and navigation controls. The terminal window shows a root user on a server, running various commands to set up Maildir. The commands include running `vagrant ssh server`, logging in as `ndbaranov`, and running `systemctl restart postfix` and `systemctl enable dovecot`. The terminal also shows the output of `mail_location` and `mail_location` commands, indicating that the mail location is set to `~/Maildir`.

```
PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering
==> server: password for you. If you're prompted for a password, please e
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2
vagrant@server:~$ su -nd baranov
su: invalid option -- 'n'
Try 'su --help' for more information.
vagrant@server:~$ su - ndbaranov
Password:
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1
[ndbaranov@server ~]$ MAIL=~/Maildir mail
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory
[ndbaranov@server ~]$ duo -i
bash: duo: command not found...
[ndbaranov@server ~]$ sudo -i
[sudo] password for ndbaranov:
[root@server ~]# doveadm mailbox list -u user
doveadm(user): Error: User doesn't exist
[root@server ~]#
```

Создание структуры Maildir

- Для пользователя созданы каталоги cur, new и tmp.

```
root@server:~  
PS D:\work\ndbaranov\vagrant> vagrant ssh server  
==> server: The machine you're attempting to SSH into is configured to use  
==> server: password-based authentication. Vagrant can't script entering the  
==> server: password for you. If you're prompted for a password, please enter  
==> server: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2  
vagrant@server:~$ su -nd baranov  
su: invalid option -- 'n'  
Try 'su --help' for more information.  
vagrant@server:~$ su - ndbaranov  
Password:  
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1  
[ndbaranov@server ~]$ MAIL=~/.Maildir mail  
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir  
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory  
[ndbaranov@server ~]$ duo -i  
bash: duo: command not found...  
[ndbaranov@server ~]$ sudo -i  
[sudo] password for ndbaranov:  
[root@server ~]# dovecadm mailbox list -u user  
doveadm(user): Error: User doesn't exist
```

Подключение к IMAP

- Через telnet установлено соединение с портом 143 и выполнена команда LOGIN.

```
root@client:~  
PS D:\work\ndbaranov\vagrant> vagrant ssh server  
==> server: The machine you're attempting to SSH into is configured to use  
==> server: password-based authentication. Vagrant can't script entering the  
==> server: password for you. If you're prompted for a password, please enter  
==> server: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2  
vagrant@server:~$ su -nd baranov  
su: invalid option -- 'n'  
Try 'su --help' for more information.  
vagrant@server:~$ su - ndbaranov  
Password:  
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1  
[ndbaranov@server ~]$ MAIL=~/.Maildir mail  
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir  
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory  
[ndbaranov@server ~]$ duo -i  
bash: duo: command not found...  
[ndbaranov@server ~]$ sudo -i  
[sudo] password for ndbaranov:  
[root@server ~]# doveadm mailbox list -u user  
doveadm(user): Error: User doesn't exist  
[root@server ~]# doveadm mailbox list -u ndbaranov  
INBOX  
[root@server ~]# exit  
logout  
[ndbaranov@server ~]$ exit  
logout  
vagrant@server:~$ exit  
logout  
PS D:\work\ndbaranov\vagrant> vagrant ssh client  
==> client: The machine you're attempting to SSH into is configured to use  
==> client: password-based authentication. Vagrant can't script entering the  
==> client: password for you. If you're prompted for a password, please enter  
==> client: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sun Sep  6 08:50:22 2026 from 10.0.2.2  
vagrant@client:~$ sudo -i  
root@client:~# dnf -y install evolution  
Last metadata expiration check: 0:12:50 ago on Sun 06 Sep 2026 09:09:47 AM UTC.  
Dependencies resolved.  
=====
```

Работа с папкой INBOX

- Команды LIST, SELECT и FETCH вывели папки и содержимое письма.

```
root@client:~# id ndbaranov
id: 'ndbaranov': no such user
root@client:~# useradd -m ndbaranov
echo 'ndbaranov:123456' | chpasswd
root@client:~# postconf -e 'inet_protocols = ipv4'
systemctl enable --now postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/postfix.service'.
root@client:~# openssl s_client -connect mail.ndbaranov.net:143 -starttls imap -crlf -quiet
Connecting to 192.168.1.1
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
verify return:1
. OK Pre-login capabilities listed, post-login capabilities have more.
a1
a1 BAD Error in IMAP command received by server.
LOGIN
LOGIN BAD Error in IMAP command received by server.
ndbaranov
* BYE Too many invalid IMAP commands.
root@client:~# openssl s_client -connect mail.ndbaranov.net:143 -starttls imap -crlf -quiet
Connecting to 192.168.1.1
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com
verify return:1
. OK Pre-login capabilities listed, post-login capabilities have more.
a1 LOGIN ndbaranov 123456
* CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS
THREAD=ORDEREDSUBJECT MULTIAPPEND URL-PARTIAL CATENATE UNSELECT CHILDREN NAMESPACE UIDPLUS LIST-EXTENDED
118NLEVEL=1 CONDSTORE QRESYNC ESEARCH ESORT SEARCHRES WITHIN CONTEXT-SEARCH LIST-STATUS BINARY MOVE SNIPPET=FUZZY
PREVIEW=FUZZY PREVIEW STATUS=SIZE SAVEDATE LITERAL+ NOTIFY SPECIAL-USE
a1 OK Logged in
a2 LIST "" ""
* LIST (\HasNoChildren) "." INBOX
a2 OK List completed (0.001 + 0.000 secs).
a3 SELECT INBOX
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Flags permitted.
* 0 EXISTS
* 0 RECENT
```

Удаление письма через IMAP

- Флаг Deleted назначен командой STORE, а EXPUNGE окончательно удалил сообщение.

```
ndbaranov@server:~  
[root@server ~]# doveadm mailbox list -u ndbaranov  
INBOX  
[root@server ~]# doveadm search -u ndbaranov mailbox INBOX ALL  
ada75421c8399d6a63400000a2dd03fa 1  
ada75421c8399d6a63400000a2dd03fa 2  
ada75421c8399d6a63400000a2dd03fa 3  
[root@server ~]# doveadm mailbox list -u ndbaranov  
INBOX  
[root@server ~]# su - ndbaranov  
Last login: Sun Sep  6 09:20:29 UTC 2026 on pts/2  
[ndbaranov@server ~]$ MAIL=~/.Maildir mail  
s-nail version v14.9.24. Type '?' for help  
/home/ndbaranov/Maildir: 3 messages 3 new  
#N 1 Super User      2026-09-06 10:02  17/664  "IMAP test 1"  
#N 2 Super User      2026-09-06 10:02  17/664  "IMAP test 2"  
#N 3 Super User      2026-09-06 10:02  17/666  "IMAP test 3"  
& q  
Held 3 messages in /home/ndbaranov/Maildir  
[ndbaranov@server ~]$ telnet mail.ndbaranov.net 110  
Trying 192.168.1.1...  
Connected to mail.ndbaranov.net.  
Escape character is '^['.  
+OK Dovecot ready.  
user ndbaranov  
+OK  
pass 123456  
+OK Logged in.  
list  
+OK 3 messages:  
1 679  
2 679  
3 681  
  
retr 1  
+OK 679 octets  
Return-Path: <root@client.ndbaranov.net>  
X-Original-To: ndbaranov@ndbaranov.net  
Delivered-To: ndbaranov@ndbaranov.net  
Received: from client.ndbaranov.net (unknown [192.168.1.31])  
by server.ndbaranov.net (Postfix) with ESMTPS id E05186007316  
for <ndbaranov@ndbaranov.net>; Sun,  6 Sep 2026 10:02:12 +0000 (UTC)  
Received: by client.ndbaranov.net (Postfix from userid 0)
```

Раздел 4

Результаты

- Dovecot хранит сообщения в Maildir и предоставляет доступ по IMAP. Проверены аутентификация, просмотр папок, чтение, установка флагов и удаление письма.

Спасибо за внимание